MOCH. LUTHFI HADIWIDJAJA

Software Engineer - Trading Systems, Web3 Infrastructure & Exchange-Adjacent Backend

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SUMMARY

Software engineer working where money movement meets software: a bank finance backend in Java and Oracle (Deposit module, API contracts, database-level reconciliation), quality engineering on enterprise platforms, and a crypto trading system I designed, built and still operate end to end. Qkuk Terminal ingests roughly 470 Binance futures pairs in real time, runs four signal engines, gates every strategy through statistical validation before it may trade, and publishes its own state to a web terminal that I wrote from scratch. Seven years in Web3 across 500+ testnets (nodes, RPC endpoints, wallets, bridges, confirmations, reorgs). I am aiming at exchange infrastructure: ledger-backed deposits and withdrawals, market data services, matching and custody operations.

CORE SKILLS

- **Backend & APIs:** Java, Spring Boot, REST API design, Node.js, .NET/C#, service layer design, idempotency and reconciliation logic, JSON/CSV storage pipelines
- **Data & accounting:** Oracle Database, PostgreSQL, MySQL, SQLite, SQL tuning, ledger-style transaction records, double-entry reasoning, audit-friendly history
- **Trading systems:** real-time market data ingestion, signal/order lifecycle, entry-stop-target modelling, backtesting harness, walk-forward and out-of-sample splits, deflated Sharpe, PBO, purged cross-validation, Binance Futures/Spot APIs
- **Blockchain / Web3:** EVM wallets and RPC, transaction lifecycle and confirmation policy, node and client operation, bridges, block explorers, on-chain data, Solidity basics
- **Frontend:** JavaScript (ES6), canvas and SVG data visualisation, responsive CSS, Flutter/Dart
- **Quality & security:** test design, defect reproduction, database-level verification, access and secret handling, restart-safe long-running services
- **Tooling:** Git, Linux, Bash, Docker, Postman, cron/supervisor-style process control, GitHub Actions and Pages

PROFESSIONAL EXPERIENCE

Software Engineer / QA Engineer - December 2024 - Present

PT. LG Sinarmas Technology Solutions | Contract | Indonesia

Delivery consultant across three client engagements, moving between development and quality assurance as each project required.

KBI NGBS - Finance Development Team

- Designed and implemented Java service-layer components against an Oracle database for the Deposit module - the path money takes when it enters the system.
- Wrote and tuned SQL, then verified results directly against the database instead of trusting application output; reconciling records is the whole job when a deposit is disputed.
- Agreed API contracts with the Deposit team and ran integration testing through to defect resolution.

CleverTap - Quality Assurance

- Tested the CleverTap customer engagement platform as partner-side QA, covering functional behaviour across the product.
- Completed vendor training and earned two partner certifications: Platform Core Functionalities and Platform Advanced Functionalities.

LG Electronics - Warehouse Systems, QA (current)

- Testing warehouse and goods-shipping systems: inbound and outbound flows, inventory movement, and the logistics records behind them.
- Trace records end to end whenever reported numbers disagree, and reproduce failures with the conditions needed to fix them.

Path in

- Joined through the Tel-U and LG CNS Java, C# and Oracle Fusion Institute Bootcamp, a selective programme run with Indonesian universities; the strongest performers were offered a place on the team.

Web3 Testnet Contributor - June 2019 - June 2024

Freelance | Remote

- Exercised pre-release builds across 500+ Web3 projects: ran nodes and clients over widely different network architectures, reproduced failures and reported defects with the exact conditions required to reproduce them.

- Worked daily with wallets, RPC endpoints, bridges, block explorers and faucets - including what confirmations, stuck transactions and chain reorganisations actually look like from the outside.
- Observed a broad range of token designs, consensus mechanisms and failure modes, including many projects that never reached mainnet.

Research Assistant, Computer Vision Division - April 2016 - April 2018
LIPI Bandung, Indonesian Institute of Sciences | Contract | Bandung

WoodID

- Built version one of an automated wood-species identification system in C as a desktop application, plus the graphical interface that let forestry staff use it without a developer present.
- Validated the system against thousands of specimens with the Ministry of Forestry and kept the instrument in service.
- Co-authored a paper published through IOP Science (ICODIS 2018) and contributed to two registered Indonesian software copyrights.

SELECTED PROJECTS

Qkuk Terminal - autonomous crypto trading system | 2025 - present | solo

- Four signal engines in Python over Binance USD-M futures, roughly 470 pairs polled on 1h, 2h and 4h timeframes. Entry, stop and target are fixed before a signal is published, so the record can be scored later without hindsight.
- Signal lifecycle from detection to resolution: watchlist queue, entry, TP/SL resolution, live win rate - every state change persisted to CSV/JSON and served by a Node.js and vanilla-JS terminal with a canvas bubble map, composite direction gauge, equity curves and per-hour win-rate analysis.
- Statistical gate before production: deflated Sharpe ratio, probability of backtest overfitting and combinatorial purged cross-validation over 700+ days of history. When out-of-sample expectancy decayed, the engine was retired rather than defended - SWING 4H and Wyckoff are gone, SCALP 2H was promoted on +0.197R out-of-sample across 3,637 trades.
- Operations I had to build to keep it honest: supervisor daemon with automatic respawn, per-process API throttling, response caching, DNS-over-HTTPS fallback when the exchange host was blocked by the local resolver, and a scheduled publishing path that writes only its own data files into a shared repository.
- A TradingView Pine Script v6 indicator mirrors the engine logic - MSS/BOS, FVG/BPR, ATR-impulse order blocks, liquidity sweeps, CISD, inducement, VWAP bands, volume profile, killzones - so a setup can be cross-checked on the chart. Disagreements between the two have caught real bugs.

SINERGI K3 - field inspection and reporting app | Flutter | client work, PT. Toba Royal Indonesia

- Offline-first K3 inspection workflow (141 Dart files): the whole inspection runs with no network, because inspectors work in lift shafts and plant rooms where there is none.
- Generates official Microsoft Word reports on the phone from nine Permenaker-aligned templates.
- Hive encrypted with an AES key held in the Android Keystore, bcrypt password hashing, custom in-app camera, backup and restore to a portable archive.

WoodID - automated wood species identification | desktop and mobile

- Local Binary Pattern and Hough Transform over images of wood surfaces; validated with the Ministry of Forestry; presented at the Forestry postgraduate programme, IPB.
- Two registered Indonesian software copyrights (ID 090683, ID 000113830); ICODIS 2018 paper through IOP Science.

EDUCATION

Bachelor of Computational Science - Telkom University, Indonesia

CERTIFICATIONS

- CleverTap Platform Core Functionalities - partner certification
- CleverTap Platform Advanced Functionalities - partner certification
- Dicoding Academy Android Developer Bootcamp - capstone, Kotlin and Jetpack Compose

ADDITIONAL

Languages: Indonesian (native), English (professional working proficiency). Open to remote full-time or contract engagement; available for a technical interview at short notice. Links: live terminal at luthfihadi78.github.io/qkuk and portfolio at luthfihadi78.github.io/portofolio.